MA2031 EndSem Solutions

A1. Using Gauss-Jordan elimination, determine the values of $\alpha \in \mathbb{R}$ so that the linear system $\quad x+y+\alpha z=1, \quad x-y-z=2, \quad 2x+y-2z=3 \quad$ has
(a) no solutions (b) a unique solution (c) infinitely many solutions. [5]

Sol: We take the first equation as the third equation. Using Gauss-Jordan elimination, we get

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 2 & 1 & -2 & 3 \\ 1 & 1 & \alpha & 1 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} \boxed{1} & 0 & -1 & 5/3 \\ 0 & \boxed{1} & 0 & -1/3 \\ 0 & 0 & \alpha+1 & -1/3 \end{array}\right]$$

(a) So, the system has no solutions when $\alpha = -1$.
(b) The system has a unique solution when $\alpha \neq -1$.
(c) The system does not have infinitely many solutions for any value of α.

A2. Determine $[A]_{E,E}$, the coordinate matrix of A with respect to the ordered basis E, where $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ and $E = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. [5]

Sol: Form the matrix P by taking the basis vectors as its columns. Then, $[A]_{E,E} = P^{-1}AP$. Now,

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Then, $P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$.

$$[A]_{E,E} = P^{-1}AP = \frac{1}{2}\begin{bmatrix} 7 & 9 & 6 \\ -1 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}.$$

Aliter: Form the matrix P by taking the basis vectors as its columns. Then, $[P|AP] \xrightarrow{\text{RREF}} [I|[A]_{E,E}]$. Now,

$$AP = \begin{bmatrix} 3 & 5 & 4 \\ 4 & 5 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

$$[P|AP] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/2 & 9/2 & 3 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 1/2 & 0 \end{array}\right].$$

Hence, $[A]_{E,E} = \frac{1}{2}\begin{bmatrix} 7 & 9 & 6 \\ -1 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix}$.

A3. The matrix $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ has two eigenvalues 1 and 3 with corresponding eigenvectors as $\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

(a) Determine the other eigenvalue and its corresponding eigenvector. [3]

(b) Using the eigenvalues and the eigenvectors, determine $A^5 \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix}$. [3]

Sol: (a) The other eigenvalue is $\text{trace}(A) - 1 - 3 = 2 - 4 = -2$.
For an eigenvector, we solve the equations $(A + 2I)v = 0$, or,
$3v_1 - v_2 + 4v_3 = 0, 3v_1 + 4v_2 - v_3 = 0, 2v_1 + v_2 + v_3 = 0$. It gives $v_1 = -v_3$, $v_2 = v_3$.
So, an eigenvector is $[-1, 1, 1]^t$.
(b) Now, $[-2, 5, 2]^t = [-1, 4, 1]^t + [-1, 1, 1]^t$.
So, $A^5[-2, 5, 2]^t = 1^5[-1, 4, 1]^t + (-2)^5[-1, 1, 1]^t = [31, -28, -31]^t$.

A4. Using Gram-Schmidt orthogonalization, compute $\text{rank}(A)$, where [4]

$$A = \begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}.$$

Sol: Taking columns as vectors, we get

$v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$, $v_3 = \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix}$, $v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. So, $\text{rank}(A) = 3$.

Aliter: Consider the row vectors to get
$v_1 = [2, -1, -1, 0]$, $v_2 = [4, 1, 7, -24]$, $v_3 = [0, 1, -1, -4] - \frac{90}{641}[4, 1, 7, -24] \neq 0$.
So, $\text{rank}(A) = 3$.

A5. Determine the best approximation of $(1, 2, 1)$ from $U = \{(a, b, c) \in \mathbb{R}^3 : a+b+c = 0\}$ [3]

Sol: The best approximation (a, b, c) of $(1, 2, 1)$ from U satisfies
(i) $(a, b, c) \in U$ and (ii) $[(1, 2, 1) - (a, b, c)] \perp U$.
A basis for U is $\{(1, -1, 0),\ (1, 0, -1)\}$.
Thus, the best approximation satisfies $(1 - a, 2 - b, 1 - c) \cdot (1, -1, 0) = 0$,
$(1 - a, 2 - b, 1 - c) \cdot (1, 0, -1) = 0$ and $a + b + c = 0$.
Solving these we find that $a = -1/3$, $b = 2/3$, $c = -1/3$.
Hence, the best approximation is $(-1/3, 2/3, -1/3)$.

A6. Find the singular value decomposition of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$. [8]

Sol: Singular values are $s_1 = 5$, $s_2 = 3$. (one may also write $s_3 = 0$.)

For $s_1^2 = 25$, $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$; For $s_2^2 = 9$, $v_2 = \frac{1}{\sqrt{18}} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$;

For $s_3^2 = 0$, $v_3 = \frac{1}{3} \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}$.

$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

$$A = P\Sigma Q^* = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{18} & -1/\sqrt{18} & 4/\sqrt{18} \\ -2/3 & -2/3 & -1/3 \end{bmatrix}.$$

B7. Let $\{v_1, \ldots, v_n\}$ be a given basis of $\mathbb{R}^{n\times 1}$. Let $A \in \mathbb{R}^{n\times n}$. Prove that A is invertible if and only if $Ax = v_i$ is consistent for each $i \in \{1, \ldots, n\}$. [3]

Sol: Let A be invertible. Then, a solution of $Ax = v_i$ is $x = A^{-1}v_i$. So, $Ax = v_i$ is consistent for each i.

Conversely, suppose for each i, $Ax = v_i$ is consistent. So, let u_i be a solution of $Ax = v_i$. Then, $A[u_1, \cdots, u_n] = [v_1, \cdots, v_n]$. Since $\{v_1, \ldots, v_n\}$ is a basis of $\mathbb{R}^{n\times 1}$, the matrix $[v_1, \cdots, v_n]$ is invertible. Thus, $A[u_1, \cdots, u_n][v_1, \cdots, v_n]^{-1} = I$. As A is a square matrix, its inverse is $[u_1, \cdots, u_n][v_1, \cdots, v_n]^{-1}$.

B8. Let $A \in \mathbb{R}^{3\times 3}$ have all its entries equal to 1. Determine a matrix P so that $P^{-1}AP$ is a diagonal matrix. [3]

Sol: $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. It has eigenvalues $3, 0, 0$.

Corresponding linearly independent eigenvectors are $[1, 1, 1]^t$, $[1, -1, 0]^t$ and $[1, 0, -1]^t$.

Hence the matrix $P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ is a diagonalizing matrix.

B9. Let A be a nonzero matrix of rank 1. What are its eigenvalues? [3]

Sol: Let $A \in \mathbb{C}^{n\times n}$. $\text{rank}(A) = 1$ implies $\text{null}(A) = n - 1$.
That is, $Ax = 0$ has $n - 1$ linearly independent solutions;
these are eigenvectors for the eigenvalue 0.
Then, the other eigenvalue is $\text{trace}(A)$.

B10. Let $A \in \mathbb{R}^{5\times 5}$. Suppose $\text{Sol}(A, 0) = \{[a, b, c, d, e]^t \in \mathbb{R}^{5\times 1} : a + d + e = 0 = b + c\}$ and $\text{rank}(A - I) = 3$. Determine whether A is diagonalizable. [3]

Sol: Geometric multiplicity of eigenvalue 1 is $\text{null}(A - 1 \cdot I) = 5 - 3 = 2$.
Geometric multiplicity of eigenvalue 0 is $\text{null}(A - 0 \cdot I) = \dim(\text{Sol}(A, 0)) = 3$.
Their sum is 5 which is the order of the matrix. Hence, A is diagonalizable.

B11. Let $A \in \mathbb{C}^{n\times n}$ be a normal matrix. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A associated with an eigenvector $v \in \mathbb{C}^{n\times 1}$. Show that $\overline{\lambda}$ is an eigenvalue of A^* associated with the eigenvector v. [3]

Sol: Let the eigenvalue of A corresponding to the eigenvector v by λ.
Then, $\|Av - \lambda v\|^2 = 0$.
Its expansion gives $0 = v^*A^*Av - \overline{\lambda}v^*Av - \lambda v^*Av + |\lambda|^2 v^*v$.
Use ${}^*A = AA^*$ and rearrange the RHS to get $\|A^*v - \overline{\lambda}v\|^2$.
Hence, $\|A^*v - \overline{\lambda}v\|^2 = 0$ so that v is an eigenvector of A^*.

Aliter: Since A is normal, there exists unitary P such that $A = PDP^*$.
Assume that λ occurs as the ith diagonal entry in D and the corresponding eigenvector v is the ith column of P. Now, $A^* = PD^*P^*$ says that $\overline{\lambda}$ is the ith diagonal entry of D^* and the corresponding eigenvector is the ith column of P, that is, v. Hence, v is an eigenvector associated with the eigenvalue $\overline{\lambda}$ of A^*.

B12. Let a, b, c, d be distinct real numbers. Let u, v, w be linearly independent vectors in $\mathbb{R}^{5\times 1}$. Determine whether the vectors $u + av$, $v + bw$, $w + cu$ and $u - v - dw$ are linearly independent or linearly dependent. [2]

Sol: Let $U = \text{span}\{u, v, w\}$. Since u, v, w are linearly independent, $\dim(U) = 3$.
If a vector in the list $u + av$, $v + bw$, $w + cu$, $u - v - dw$ is repeated, then they are linearly dependent. Otherwise, they are four vectors in U, which is a 3-dimensional subspace of $\mathbb{C}^{5\times 1}$; so they are linearly dependent.
In any case, the given vectors are linearly dependent.

B13. Construct a matrix $A \in \mathbb{R}^{4\times 4}$ where no column is a linear combination of any two other columns, but $\det(A) = 0$. [2]

Sol: Take $A = [e_1,\ e_2,\ e_3,\ e_1 + e_2 + e_3]$. Now, no column is a linear combination of other two columns. But $\det(A) = 0$.

B14. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Does there exist an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix? Justify your answer. [2]

Sol: Suppose such a matrix P exists. The eigenvalues of A are $1, 1, 1, 1$. So, $P^{-1}AP = I$. Now, $P^{-1}(A - I)P = P^{-1}AP - I = 0$ has rank 0. But $A - I$ has rank 1. This contradicts the fact that the ranks of two similar matrices are equal.
Hence, there does not exist such a matrix P.

B15. Let A be a square matrix with characteristic polynomial $(t-1)^2(t-2)^3(t-3)^4(t-4)$. Determine the rank of A. [2]

Sol: $\det(A)$ = product of its eigenvalues $= 1^2 \cdot 2^3 \cdot 3^4 \cdot 4 \neq 0$. So, A is invertible.
Hence, A has rank = its order $= 2 + 3 + 4 + 1 = 10$.

B16. Let $V = \left\{ (a,b,c) \in \mathbb{R}^3 : \begin{bmatrix} 7 & a & b \\ 0 & 2 & c \\ 0 & 0 & 3 \end{bmatrix} \text{ is diagonalizable} \right\}$.

Determine whether V is a subspace of $\mathbb{R}^3$. [2]

Sol: The given matrix there is upper triangular; its diagonal entries are its eigenvalues. They are distinct. So, the matrix is diagonalizable for all values of $a, b.c.$ Hence, $V = \mathbb{R}^3$. It is a subspace of $\mathbb{R}^3$.

B17. Let $A \in \mathbb{C}^{10\times 10}$ have the characteristic polynomial $(t-1)^7(t-2)^2(t-3)$.
If $\text{rank}(A-I) = 5$, then determine whether A is diagonalizable or not. [2]

Sol: The geometric multiplicity of the eigenvalue 1 is $\text{null}(A-I) = 10-5 = 5$.
The algebraic multiplicity of 1 is $7 > 5$. So, A is not diagonalizable.

B18. Let $A \in \mathbb{R}^{3\times 3}$, $A \neq 0$ but $A^2 = 0$. Determine whether A is diagonalizable. [2]

Sol: If $\lambda \in \sigma(A)$, then $Av = \lambda v$ implies $0 = A^2 v = \lambda^2 v$ implies $\lambda = 0$.
That is, the only eigenvalue is 0.
If A is diagonalizable, then $A = P^{-1}0P = 0$, a contradiction.
Hence, A is not diagonalizable.

$A = PDP^{-1}$

$A^2 = PDP^{-1}PDP^{-1}$

$0 = PD^2P^{-1}$

$\Rightarrow D^2 = 0$

$\Rightarrow \lambda_i = 0 \Rightarrow D = 0$

$\Rightarrow$ contradiction

A1. Using Gauss-Jordan elimination, determine the values of $\alpha \in \mathbb{R}$ so that the linear system $x + y + \alpha z = 1, \quad x - y - z = 2, \quad 2x + y - 2z = 3$ has
(a) no solutions (b) a unique solution (c) infinitely many solutions. [5]

$$\begin{bmatrix} 1 & 1 & \alpha & 1 \\ 1 & -1 & -1 & 2 \\ 2 & 1 & -2 & 3 \end{bmatrix} \xrightarrow[E_{-2}[3,1]]{E_{-1}[2,1]} \begin{bmatrix} 1 & 1 & \alpha & 1 \\ 0 & -2 & -(\alpha+1) & 1 \\ 0 & -1 & -2(\alpha+1) & 1 \end{bmatrix}$$

$$\xrightarrow{E_{1/2}[2]} \begin{bmatrix} 1 & 1 & \alpha & 1 \\ 0 & 1 & \frac{\alpha+1}{2} & -1/2 \\ 0 & -1 & -2(\alpha+1) & 1 \end{bmatrix}$$

$$\xrightarrow[E_{1}[3,2]]{E_{-1}[1,2]} \begin{bmatrix} 1 & 0 & \frac{\alpha-1}{2} & \frac{3}{2} \\ 0 & 1 & \frac{\alpha+1}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{-3(\alpha+1)}{2} & 1/2 \end{bmatrix}$$

$$\xrightarrow{E_{\frac{2}{3(\alpha+1)}}[3]} \begin{bmatrix} 1 & 0 & \frac{\alpha-1}{2} & \frac{3}{2} \\ 0 & 1 & \frac{\alpha+1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{3(\alpha+1)} \end{bmatrix} \Rightarrow \boxed{\alpha \neq -1}$$

$$\xrightarrow[E_{-\frac{\alpha+1}{2}}[2,3]]{E_{-\frac{\alpha-1}{2}}[1,3]} \begin{bmatrix} 1 & 0 & 0 & \frac{\alpha+2}{3(\alpha+1)} \\ 0 & 1 & 0 & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3(\alpha+1)} \end{bmatrix}$$

(a) for NO SOLUTION: $\text{Rank}[A|b] \neq \text{Rank}[A]$

$\Rightarrow \alpha = -1$

(b) for 1 SOLUTION: $\text{Rank}[A|b] = \text{Rank}[A] = n \Rightarrow \alpha \in \mathbb{R} - \{-1\}$

(c) for ∞ SOLUTIONS: $\text{Rank}[A|b] = \text{Rank}[A] < n \Rightarrow$ NO α

A6. Find the singular value decomposition of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}_{2\times 3}$.

$$AA^* = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

$$\sigma(AA^*) = \{\lambda : (\lambda-17)^2 = 8^2\} = \{9, 25\} \quad \text{rank} = 2$$

$$\Rightarrow s_1 = 3, \; s_2 = 5 \Rightarrow \Sigma = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix}$$

$$U \in C^{2\times 2} \quad V \in C^{3\times 3}$$

$$AA^* u_i = \begin{cases} s_i^2 u_i & 1 \le i \le 2 \\ 0 & \text{nowhere} \end{cases}$$

$$\Rightarrow \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix} u_1 = 9u_1 \quad u_1 \Rightarrow N(A-9I) = N\left(\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}\right)$$

$$= \text{span}([1, -1]^t)$$

$$\Rightarrow u_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix} u_2 = 25u_2 \Rightarrow u_2 \Rightarrow N(A-25I) = \text{span}([1,1]^t)$$

$$\Rightarrow u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$v_i = \frac{A^* u_i}{s_i}$$

$$v_1 = \frac{\begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}}{3} = \frac{\begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 4/\sqrt{2} \end{bmatrix}}{3} = \begin{bmatrix} 1/\sqrt{18} \\ -1/\sqrt{18} \\ 4/\sqrt{18} \end{bmatrix}$$

$$v_2 = \frac{\begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}}{5} = \frac{\begin{bmatrix} 5/\sqrt{2} \\ 5/\sqrt{2} \\ 0 \end{bmatrix}}{5} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$v_3 : A^*A v_3 = 0$

$$A^*A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 13 & 12 & 2 \\ 12 & 13 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

$13v_1 + 12v_2 + 2v_3 = 0 \Rightarrow 25v_1 + 50v_3 = 0$

$12v_1 + 13v_2 - 2v_3 = 0 \Rightarrow v_1 = -2v_3$

$v_1 - v_2 + 4v_3 = 0 \Rightarrow v_2 = v_1 + 4v_3 = 2v_3$

$\Rightarrow \text{span}[-2, 2, 1]$

$v_3 = [-2/3, 2/3, 1/3]^t$

$$V = \begin{bmatrix} 1/\sqrt{18} & 1/\sqrt{2} & -2/3 \\ -1/\sqrt{18} & 1/\sqrt{2} & 2/3 \\ 4/\sqrt{18} & 0 & 1/3 \end{bmatrix}$$

$A = U\Sigma V^*$

A2. Determine $[A]_{E,E}$, the coordinate matrix of A with respect to the ordered basis E,

where $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ and $E = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. [5]

$[A]_{E,E} = E^{-1}AE$

$AE = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 & 4 \\ 4 & 5 & 3 \\ 0 & 1 & 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 3 & 5 & 4 \\ 1 & 0 & 1 & 4 & 5 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array}\right]$$

$\downarrow E_{+}[2,1]$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 3 & 5 & 4 \\ 0 & -1 & 1 & 1 & 0 & -1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array}\right]$$

$\downarrow E_1[1,2]$
$E_1[3,2]$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 4 & 5 & 3 \\ 0 & -1 & 1 & 1 & 0 & -1 \\ 0 & 0 & 2 & 1 & 1 & 0 \end{array}\right]$$

$\downarrow E_{-1}[2]$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 4 & 5 & 3 \end{array}\right]$$

$$\left[\begin{array}{cccccc} 0 & 1 & -1 & -1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 & 0 \end{array}\right]$$

$$\downarrow E_{-1/2}[1,3] \quad E_{1/2}[2,3]$$

$$\left[\begin{array}{cccccc} 1 & 0 & 0 & 7/2 & 9/2 & 3 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1 \\ 0 & 0 & 2 & 1 & 1 & 0 \end{array}\right]$$

$$\downarrow E_{1/2}[3]$$

$$\left[\begin{array}{cccccc} 1 & 0 & 0 & 7/2 & 9/2 & 3 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1 \\ 0 & 0 & 1 & 1/2 & 1/2 & 0 \end{array}\right]$$

$$[A]_{EE} = \left[\begin{array}{ccc} 7/2 & 9/2 & 3 \\ -1/2 & 1/2 & 1 \\ 1/2 & 1/2 & 0 \end{array}\right]$$

A3. The matrix $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ has two eigenvalues 1 and 3 with corresponding eigenvectors as $\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

(a) Determine the other eigenvalue and its corresponding eigenvector. [3]

(b) Using the eigenvalues and the eigenvectors, determine $A^5 \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix}$. [3]

(a)

$$tr(A) = 2 = \lambda_1 + \lambda_2 + \lambda_3 = 1 + 3 + \lambda_3 \Rightarrow \lambda_3 = -2$$

$$\begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = -2 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\Rightarrow x_1 - x_2 + 4x_3 = -2x_1 \Rightarrow 3x_1 - x_2 + 4x_3 = 0$

$3x_1 + 2x_2 - x_3 = -2x_2 \Rightarrow 3x_1 + 4x_2 - x_3 = 0$

$2x_1 + x_2 - x_3 = -2x_3 \Rightarrow 2x_1 + x_2 + x_3 = 0$

$$\Rightarrow \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

(b) $A\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} = 1\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ $\quad A\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = -2\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$A^2\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ $\quad A^2\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = 4\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$A^5\begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ $\quad A^5\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = -32\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$$A^5\begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 31 \\ -28 \\ -31 \end{bmatrix}$$

A4. Using Gram-Schmidt orthogonalization, compute rank(A), where

$$A = \begin{bmatrix} 2 & -1 & -1 & 0 \\ 1 & 0 & 1 & -4 \\ 0 & 1 & -1 & -4 \end{bmatrix}.$$

$\theta_1 = [2 \ 1 \ 0]^t$ $\quad \theta_2 = [-1 \ 0 \ 1]^t$

$\theta_3 = [-1 \ 1 \ -1]^t$ $\quad \theta_4 = [0 \ -4 \ -4]^t$

$u_1 = [2 \ 1 \ 0]^t$

$u_2 = v_2 - \frac{\langle v_2 u_1 \rangle}{\langle u_1 \cdot u_1 \rangle} u_1 = [-1 \ 0 \ 1]^t - \frac{(-2)}{5}[2 \ 1 \ 0]^t$

$= \left[\frac{-1}{5} \ \frac{2}{5} \ 1\right]^t$

$u_3 = v_3 - \frac{\langle v_3 u_1 \rangle}{\langle u_1 u_1 \rangle} v_1 - \frac{\langle v_3 u_2 \rangle}{\langle u_2 u_2 \rangle} v_2$

$= [-1 \ 1 \ -1]^t - \frac{(-1)}{5}[2 \ 1 \ 0]^t - \frac{(-2/5)}{6/5}\left[\frac{-1}{5} \ \frac{2}{5} \ 1\right]^t$

$= \left[\frac{-2}{3} \ \frac{4}{3} \ \frac{-2}{3}\right]^t \qquad 4/9 + 16/9 + 4/9$

$u_4 = v_4 - \frac{\langle v_4 u_1 \rangle}{\langle u_1 u_1 \rangle} u_1 - \frac{\langle v_4 u_2 \rangle}{\langle u_2 u_2 \rangle} u_2 - \frac{\langle v_4 u_3 \rangle}{\langle u_3 u_3 \rangle} u_3$

$= [0 \ 4 \ -4]^t - \frac{(-4)}{5}[2 \ 1 \ 0]^t - \frac{(-28/5)}{6/5}[-1/5 \ 2/5 \ 1]^t$

$- \frac{(-8/3)}{(8/3)}[-2/3 \ 4/3 \ -2/3]^t$

$= [0, 0, 0]^t \Rightarrow \text{rank} = 3$

A5. Determine the best approximation of $(1, 2, 1)$ from $U = \{(a, b, c) \in \mathbb{R}^3 : a+b+c = 0\}$. [3]

$(a, b, -a-b) = (1, 0, -1), (0, 1, -1)$

$v_1 = (1/\sqrt{2}, 0, -1/\sqrt{2}) = w_1$

$v_2 = (0, 1, -1) - 1/\sqrt{2}\,(1/\sqrt{2}, 0, -1/\sqrt{2}) = (-\frac{1}{2}, 1, -\frac{1}{2})$

$w_2 = \left[\frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}\right]$

$y = \sum_i \langle x, w_i \rangle w_i = \frac{2}{\sqrt{6}}\left[-1/\sqrt{6}, 2/\sqrt{6}, -1/\sqrt{6}\right]$

$= \left[\frac{-1}{3}, \frac{2}{3}, \frac{-1}{3}\right]$

B7. Let $\{v_1, \ldots, v_n\}$ be a given basis of $\mathbb{R}^{n\times 1}$. Let $A \in \mathbb{R}^{n\times n}$. Prove that A is invertible if and only if $Ax = v_i$ is consistent for each $i \in \{1, \ldots, n\}$. [3]

If $Ax = v_i$ is consistent:

columns with vectors $v_i \to I$

$\Rightarrow$ columns of Ax_i span entire space and is linearly independent

$A[x_1\ x_2 \ldots x_n] = [v_1\ v_2 \ldots v_n]$

$\Rightarrow A[x_1\ x_2 \ldots x_n][v_1\ v_2 \ldots v_n]^{-1} = I$

($[x_1\ x_2 \ldots x_n]$: A inverse)

If A is invertible:

$Ax_i = v_i \Rightarrow x_i = A^{-1}v_i \Rightarrow$ consistent

B8. Let $A \in \mathbb{R}^{3\times 3}$ have all its entries equal to 1. Determine a matrix P so that $P^{-1}AP$ is a diagonal matrix. [3]

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} (\lambda-1) & -1 & -1 \\ -1 & \lambda-1 & -1 \\ -1 & -1 & \lambda-1 \end{vmatrix} = \lambda^2(\lambda-3)$$

$\Rightarrow 0, 0, 3 \rightarrow$ eigenvalues

eigenvectors: $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$\Rightarrow P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 1 \end{bmatrix} \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

B9. Let A be a nonzero matrix of rank 1. What are its eigenvalues?

let $A \in \mathbb{C}^{n\times n} \rightarrow \text{null}(A) = n-1$

$Ax = 0$ has $(n-1)$ independent Sol's

$\Rightarrow$ 0 is eigen value $(n-1)$ times

the one no zero eigen value is $\text{tr}(A)$

B10. Let $A \in \mathbb{R}^{5\times 5}$. Suppose $\text{Sol}(A, 0) = \{[a, b, c, d, e]^t \in \mathbb{R}^{5\times 1} : a + d + e = 0 = b + c\}$ and $\text{rank}(A - I) = 3$. Determine whether A is diagonalizable. [3]

$Ax = 0$

$a, b, -b, d, -a-d = [1,0,0,0,-1]^t, [0,1,-1,0,0]^t,$

$[0,0,0,1,-1]^t$

Geometric multiplicity of eigenvalue 0:

$\dim(\text{Null}(A - 0 \cdot I)) = 3$

Geometric multiplicity of eigenvalue 1:

$\dim(\text{Null}(A - I)) = 2$

Sum $= 5 \Rightarrow A$ is diagonalizable

B11. Let $A \in \mathbb{C}^{n \times n}$ be a normal matrix. Let $\lambda \in \mathbb{C}$ be an eigenvalue of A associated with an eigenvector $v \in \mathbb{C}^{n \times 1}$. Show that $\overline{\lambda}$ is an eigenvalue of A^* associated with the eigenvector v. [3]

$A^*A = AA^*$

$Av = \lambda v$ ← to prove $A^*v = \overline{\lambda}v$

$Av - \lambda v = 0$ → $v^*A^* = \overline{\lambda}v^*$

$(v^*A^* - \overline{\lambda}v^*)(Av - \lambda v) = 0$

$v^*A^*Av - v^*A^*\lambda v - \overline{\lambda}v^*Av + \lambda\overline{\lambda}v^*v = 0$

$\Rightarrow v^*AA^*v - v^*A^*\lambda v - \overline{\lambda}v^*Av + \lambda\overline{\lambda}v^*v = 0$

$\Rightarrow v^*A[A^*v - \overline{\lambda}v] - \lambda v^*[A^*v - \overline{\lambda}v] = 0$

$\Rightarrow (v^*A - \lambda v^*)[A^*v - \overline{\lambda}v] = 0$

$\Rightarrow \overline{\|A^*v - \overline{\lambda}v\|^2} = 0$

$\Rightarrow \overline{\lambda}$ is eigenvalue of A^*

B12. Let a, b, c, d be distinct real numbers. Let u, v, w be linearly independent vectors in $\mathbb{R}^{5 \times 1}$. Determine whether the vectors $u + av$, $v + bw$, $w + cu$ and $u - v - dw$ are linearly independent or linearly dependent. [2]

$\alpha(u + av) + \beta(v + bw) + \gamma(w + cu) + \delta(u - v - dw) = 0$

$\Rightarrow u(\alpha + c\delta + \xi) + v(a\alpha + \beta - \xi) + w(b\beta + \delta - d\xi) = 0$

$\Rightarrow \alpha + c\delta + \xi = 0$

$a\alpha + \beta - \xi = 0$ 4 variables $\Rightarrow$ 3 equations

$b\beta + \delta - d\xi = 0$ $\Rightarrow$ linearly dependent

B13. Construct a matrix $A \in \mathbb{R}^{4\times 4}$ where no column is a linear combination of any two other columns, but $\det(A) = 0$. [2]

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

B14. Let $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Does there exist an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix? Justify your answer. [2]

$$|\lambda I - A| = \begin{vmatrix} \lambda-1 & -1 & 0 & 0 \\ 0 & \lambda-1 & 0 & 0 \\ 0 & 0 & \lambda-1 & 0 \\ 0 & 0 & 0 & \lambda-1 \end{vmatrix} = (\lambda-1)^4$$

$Ax = x \Rightarrow x_1 + x_2 = x_1$

$x_2 = x_2 \Rightarrow x_2 = 0 \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$x_3 = x_3$

$x_4 = x_4$ $\Rightarrow$ Not diagonizable

B15. Let A be a square matrix with characteristic polynomial $(t-1)^2(t-2)^3(t-3)^4(t-4)$. Determine the rank of A. [2]

$|A| \neq 0 \Rightarrow$ rank $= 2+3+4+1 = 10$

B16. Let $V = \left\{ (a,b,c) \in \mathbb{R}^3 : \begin{bmatrix} 7 & a & b \\ 0 & 2 & c \\ 0 & 0 & 3 \end{bmatrix} \text{ is diagonalizable} \right\}$.
Determine whether V is a subspace of $\mathbb{R}^3$.

upper $\Delta \Rightarrow$ eigen values are diagonal entries
$\Rightarrow$ diagonalizable for all a,b,c
$\rightarrow$ subspace of $\mathbb{R}^3$

B17. Let $A \in \mathbb{C}^{10\times 10}$ have the characteristic polynomial $(t-1)^7(t-2)^2(t-3)$. If $\text{rank}(A-I) = 5$, then determine whether A is diagonalizable or not. [2]

Geometric multiplicity of $\lambda = 1$
$\Rightarrow \dim(\text{null}(A-I)) = 5$
Algebraic multiplicity of $\lambda = 1 \rightarrow 7$
$\Rightarrow$ not diagonalizable

B18. Let $A \in \mathbb{R}^{3\times 3}$, $A \neq 0$ but $A^2 = 0$. Determine whether A is diagonalizable.

$Av = \lambda v \Rightarrow A^2 v = A\lambda v = \lambda(Av) = \lambda^2 v$
$\Rightarrow 0 = \lambda^2 v \Rightarrow \lambda = 0 \Rightarrow D = 0$
$A = P^{-1}DP = 0$
But since $A \neq 0$, A isn't diagonalizable

Indian Institute of Technology Madras

Department of Mathematics

MA2031-Linear Algebra for Engineers

End Semester Examination

Time : 2 PM-5 PM **Date : May 01, 2023** **Maximum Marks : 60**

INSTRUCTIONS :

- **This question paper contains two parts. Part A and Part B.**
- **Question numbers 1 to 6 are in Part A. Answer them in the answer script marked A.**
- **Question numbers 7 to 12 are in Part B. Answer them in the answer script marked B.**
- **Please write your name, roll number and teacher's name on the top page of both the answer scripts A and B.**

Part A - Answer all the six questions - 26 Marks

1. Let $W = \left\{ A \in \mathbb{R}^{2\times 2} \mid A \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A \right\}$.

   (a) Show that W is a subspace of $\mathbb{R}^{2\times 2}$. [1]

   (b) Find a proper subset $S \subset W$ such that $W = \text{Span}\{S\}$. [2]

   (c) Find a basis for W. [1]

   (d) Determine the dimension of W. [1]

2. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear map defined by $T(x, y, z) = (x + 2y + 3z, 4x + 5y + 6z)$. Determine the adjoint T^*. [3]

3. Consider $\mathbb{C}^3$ with the usual inner product. Define $T : \mathbb{C}^3 \to \mathbb{C}$ by $T(x, y, z) = 3x+iy+5z$. Find a vector $w \in \mathbb{C}^3$ such that $T(v) = \langle v, w \rangle$ for each $v \in \mathbb{C}^3$. [3]

4. Let $\mathbb{R}_2([0, 1])$ be the vector space of all real polynomial functions of degree atmost 2 defined on $[0, 1]$ with inner product given by $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$.

   Let $T : \mathbb{R}_2([0, 1]) \to \mathbb{R}_2([0, 1])$ be a linear map defined by $T(a_0 + a_1t + a_2t^2) = a_1t$, and $B = \{1, t, t^2\}$.

(a) Compute $\langle T(1), t\rangle$. Use this to conclude that $T \neq T^*$. [1]

(b) Show that $M_{BB}(T) = (M_{BB}(T))^*$. [2]

(c) Although $M_{BB}(T)$ is hermitian, $T \neq T^*$. Explain why is this not a contradiction. [2]

5. Consider $W = \{(x, y) \in \mathbb{R}^2 \mid 4x + 3y = 0\}$ and let $x = (2, 4) \in \mathbb{R}^2$.

(a) Find the best approximation to x from W. [3]

(b) Write $x = a + b$, where $a \in W$ and $b \in \mathbb{R}^2$ such that $\langle a, b\rangle = 0$. [2]

6. Justify why the following matrices are diagonalizable or not.

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ [2]

(b) $\begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$ [2]

(c) A^*A, for any $A \in \mathbb{C}^{m\times n}$. [1]

Part B-Answer all the six questions - 34 Marks

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7. Consider $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$.

(a) Determine the spectrum $\sigma(A)$. [2]

(b) Determine all the Eigenspaces. [4]

8. Let $\mathbb{R}_2([-1, 1])$ be the vector space of all real polynomial functions of degree atmost 2 defined on $[-1, 1]$.

(a) Convert the basis $\{1, t, t^2\}$ to an orthonormal basis with respect to the inner product $\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)dt$. [3]

(b) Using part (a), find a polynomial $q \in \mathbb{R}_2([-1, 1])$ such that $\int_{-1}^{1} p(t)q(t)dt = p(0)$ for all $p \in \mathbb{R}_2([-1, 1])$. [2]

9. Find an upper triangular matrix which is similar to the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$. [6]

10. Obtain the QR decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. [6]

11. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear map defined by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Determine the range and null space of T. [4]

(b) Is T an isomorphism? Justify the answer. [2]

12. Let A be a 3×3 real matrix satisfying $A^2 = 2A+3I$, where I is the 3×3 identity matrix, and the determinant of A is -9.

(a) Find the minimal polynomial of the matrix A. [3]

(b) Find the trace of the matrix A. [2]

1 Let $W = \left\{ A \in \mathbb{R}^{2\times 2} \mid A\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A \right\}$.

(a) Show that W is a subspace of $\mathbb{R}^{2\times 2}$. [1]

(b) Find a proper subset $S \subset W$ such that $W = \text{Span}\{S\}$. [2]

(c) Find a basis for W. [1]

(d) Determine the dimension of W. [1]

(a)

let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \begin{bmatrix} a+b & a+b \\ c+d & c+d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix}$

$\Rightarrow b = c \; ; \; a = d \Rightarrow A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$

let $A_1 = \begin{bmatrix} a_1 & b_1 \\ b_1 & a_1 \end{bmatrix} \quad A_2 = \begin{bmatrix} a_2 & b_2 \\ b_2 & a_2 \end{bmatrix}$

$\Rightarrow A_1 + \alpha A_2 = \begin{bmatrix} a_1+\alpha a_2 & b_1+\alpha b_2 \\ b_1+\alpha b_2 & a_1+\alpha a_2 \end{bmatrix} \in W$

$\Rightarrow W$ is a subspace

(b) $A = a\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$\Rightarrow S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$

(c) the set $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ forms a basis for W

(d) 2

2. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear map defined by $T(x, y, z) = (x + 2y + 3z, 4x + 5y + 6z)$. Determine the adjoint T^*. [3]

$$T(x,y,z) = (x+2y+3z, 4x+5y+6z)$$

$$\Rightarrow T \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+2y+3z \\ 4x+5y+6z \end{bmatrix}$$

$$\Rightarrow T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$T^* = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$T^*(x,y) = (x+4y, 2x+5y, 3x+6y)$$

3. Consider $\mathbb{C}^3$ with the usual inner product. Define $T : \mathbb{C}^3 \to \mathbb{C}$ by $T(x, y, z) = 3x+iy+5z$. Find a vector $w \in \mathbb{C}^3$ such that $T(v) = \langle v, w \rangle$ for each $v \in \mathbb{C}^3$. [3]

$$T = [3 \quad i \quad 5]$$

$$\text{let } w = [w_1 \quad w_2 \quad w_3]$$

$$T(v) = 3v_1 + iv_2 + 5v_3 = v_1 w_1 + v_2 w_2 + v_3 w_3$$

$$\Rightarrow w = [3 \quad i \quad 5]$$

4. Let $\mathbb{R}_2([0, 1])$ be the vector space of all real polynomial functions of degree atmost 2 defined on $[0, 1]$ with inner product given by $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$.
Let $T : \mathbb{R}_2([0, 1]) \to \mathbb{R}_2([0, 1])$ be a linear map defined by $T(a_0 + a_1 t + a_2 t^2) = a_1 t$, and $B = \{1, t, t^2\}$.

(a) Compute $\langle T(1), t \rangle$. Use this to conclude that $T \neq T^*$. [1]

(b) Show that $M_{BB}(T) = (M_{BB}(T))^*$. [2]

(c) Although $M_{BB}(T)$ is hermitian, $T \neq T^*$. Explain why is this not a contradiction. [2]

(a) $T(1) = 0 \quad (a_0 = 1, a_1 = 0, a_2 = 0)$

$\Rightarrow \langle T(1), t \rangle = \int_0^1 0 \cdot t \, dt = 0$

Since $T(1) = 0 \neq t \Rightarrow T \neq T^*$

(b) $M_{BB}(T)[i,j] = [T(b_i), b_j]$

5. Consider $W = \{(x, y) \in \mathbb{R}^2 \mid 4x + 3y = 0\}$ and let $x = (2, 4) \in \mathbb{R}^2$.

(a) Find the best approximation to x from W. [3]

(b) Write $x = a + b$, where $a \in W$ and $b \in \mathbb{R}^2$ such that $\langle a, b \rangle = 0$. [2]

$W = \text{span}\{(1, -4/3)\}$

$v_1 = \left(\frac{3}{5}, \frac{-4}{5}\right)$

$\Rightarrow$ best approximation $= \sum_1 \langle x_i v_i \rangle v_i$

$= (-2)\left(\frac{3}{5}, \frac{-4}{5}\right)$

$= \left(\frac{-6}{5}, \frac{8}{5}\right)$

(b) $x = \underbrace{\left(\frac{-6}{5}, \frac{8}{5}\right)}_{a} + \underbrace{\left(\frac{16}{5}, \frac{12}{5}\right)}_{b}$

6. Justify why the following matrices are diagonalizable or not.

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ [2]

(b) $\begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$ [2]

(c) A^*A, for any $A \in \mathbb{C}^{m \times n}$. [1]

(a) upper triangular with distinct eigenvalues

$\Rightarrow$ diagonalizable

(b) eigenvalues = 2, 2, 4

$$2x_1 + 4x_2 + 6x_3 = 2x_1 \Rightarrow x_2 = 0 \Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_2 + 2x_3 = 2x_2 \Rightarrow x_3 = 0$$

$$4x_3 = 2x_3$$

Not diagonalizable as we dont have 3 independent eigenvectors

(c) $(A^*A)^* = A^*A \Rightarrow A^*A$ is hermitian

Let $P = A^*A$

By schur's triangularization there exists a unitary matrix B that satisfies $Q = BPB^*$ and Q is upper triangular

$$\rightarrow Q^* = BP^*B^* = BPB^* = Q$$

$\Rightarrow$ Q is a diagonal matrix with real entries

$\Rightarrow Q = BPB^{-1} \Rightarrow$ P is diagonalizable

7. Consider $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$.

(a) Determine the spectrum $\sigma(A)$. [2]

(b) Determine all the Eigenspaces. [4]

$$\begin{vmatrix} \lambda-3 & -1 & 1 \\ -2 & \lambda-2 & 1 \\ & & \end{vmatrix} = 0 \Rightarrow (\lambda-3)[\lambda^2-2\lambda+2] + 1[-2\lambda+2] + 1[4+2\lambda-4]$$

$$\begin{vmatrix} -2 & -2 & \lambda \end{vmatrix}$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 8\lambda - 6 - 2\lambda + 2 + 2\lambda = 0$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\Rightarrow \lambda = 1, 2, 2$$

$$\sigma(A) = \{1, 2, 2\}$$

(b) $$\begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Rightarrow 2x_1 + x_2 - x_3 = 0$$
$$\Rightarrow 2x_1 + x_2 - x_3 = 0$$
$$2x_1 + 2x_2 - x_3 = 0$$

$$\Rightarrow x_2 = 0, \quad x_3 = 2x_1$$

$$\Rightarrow \text{eigenspace: span}\{[1\ 0\ 2]^t\}$$

$$\begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\Rightarrow x_1 + x_2 - x_3 = 0$$
$$\Rightarrow 2x_1 - x_3 = 0$$
$$2x_1 + 2x_2 - 2x_3 = 0$$

$$\Rightarrow x_3 = 2x_1 \ ; \ x_2 = x_1$$

$$\text{eigen space} = \text{span}\{[1, 1, 2]^t\}$$

9. Find an upper triangular matrix which is similar to the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$. [6]

$$\begin{vmatrix} \lambda & -1 & 0 \\ -2 & \lambda+2 & -2 \\ -2 & 3 & \lambda-2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda[\lambda^2 - 4 + 6] + 1[-2\lambda + 4 - 4] = 0$$

$$\Rightarrow \lambda[\lambda^2 + 2] - 2\lambda = 0 \Rightarrow \lambda^3 = 0$$

$$\Rightarrow \lambda = 0$$

eigenvector: $x_2 = 0$, $x_1 - x_2 + x_3 = 0 \Rightarrow \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$2x_1 - 2x_2 + x_3 = 0$

$u_1 = [1\ 0\ -1]^t$

$\Rightarrow P_1 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$

$$P_1^* A P_1 = \begin{bmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -\sqrt{2} & 2\sqrt{2} & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & \sqrt{2} \\ 2 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -2 & 2\sqrt{2} \\ 0 & 2 & -\sqrt{2} \\ 0 & 2\sqrt{2} & -2 \end{bmatrix}$$

$A_1 = \begin{bmatrix} 2 & -\sqrt{2} \\ 2\sqrt{2} & -2 \end{bmatrix} \Rightarrow \sigma(A_1) = \{0, 0\}$

eigenvector $= \begin{bmatrix} 1 \\ \sqrt{2} \end{bmatrix}$

$\Rightarrow P_2 = \begin{bmatrix} 1/\sqrt{3} & -\sqrt{2}/\sqrt{3} \\ \sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$

$$\Rightarrow \quad P_2^* A_1 P_2 = \begin{bmatrix} 1/\sqrt{3} & \sqrt{2}/\sqrt{3} \\ -\sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 2 & -\sqrt{2} \\ 2\sqrt{2} & -2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & -\sqrt{2}/\sqrt{3} \\ \sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} 2\sqrt{3} & -\sqrt{6} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{3} & -\sqrt{2}/\sqrt{3} \\ \sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -3\sqrt{2} \\ 0 & 0 \end{bmatrix}$$

$$P = P_1 \begin{bmatrix} 1 & 0 \\ 0 & P_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{3} & -\sqrt{2}/\sqrt{3} \\ 0 & \sqrt{2}/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \end{bmatrix} \quad B =$$

10. Obtain the QR decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. [6]

$$u_1 = [1 \ 1 \ 0]^t \quad u_2 = [1 \ 0 \ 1]^t \quad u_3 = [0 \ 1 \ 1]^t$$

$$v_1 = u_1 = [1 \ 1 \ 0]$$

$$v_2 = u_2 - \frac{\langle u_2 \, v_1 \rangle}{\langle v_1 \, v_1 \rangle} v_1 = [1 \ 0 \ 1]^t - \frac{1}{2}[1 \ 1 \ 0]^t$$

$$= [1/2 \ -1/2 \ 1]^t$$

$$v_3 = u_3 - \frac{\langle u_3 \, v_1 \rangle}{\langle v_1 \, v_1 \rangle} v_1 - \frac{\langle u_3 \, v_2 \rangle}{\langle v_2 \, v_2 \rangle} v_2$$

$$= [0\ 1\ 1]^t - \frac{1}{2}[1\ 1\ 0]^t - \frac{1/2}{3/2}[1/2\ -1/2\ 1]^t$$

$$= \left[-\frac{2}{3}\ \frac{2}{3}\ \frac{2}{3}\right]^t$$

$$w_1 = [1/\sqrt{2}\ \ 1/\sqrt{2}\ \ 0]^t$$

$$w_2 = [1/\sqrt{6}\ \ -1/\sqrt{6}\ \ 2/\sqrt{6}]^t$$

$$w_3 = \left[\frac{-1}{\sqrt{3}}\ \frac{1}{\sqrt{3}}\ \frac{1}{\sqrt{3}}\right]^t$$

$$\Rightarrow Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

$$R = Q^* A = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{6} & -1/\sqrt{6} & 2/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

11. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear map defined by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Determine the range and null space of T. [4]

(b) Is T an isomorphism? Justify the answer. [2]

(a) $T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$

RREF of T: $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow[E_{-2}[3,1]]{E_{-1}[2,1]} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$\downarrow E_{-1}[3,2]$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

range of T: $[1\ 1\ 2]^t, [0\ 1\ 1]^t$

Null space of T: $[1\ 1\ -1]^t$

(b) The map isn't an isomorphism since

$Ax = 0$ doesn't yield only $\vec{0}$ (not injective)

12. Let A be a 3×3 real matrix satisfying $A^2 = 2A + 3I$, where I is the 3×3 identity matrix, and the determinant of A is -9.

(a) Find the minimal polynomial of the matrix A. [3]

(b) Find the trace of the matrix A. [2]

$A^2 - 2A - 3I = 0 \qquad \lambda_1 = -1,\ \lambda_2 = 3$

$\Rightarrow$ This is the minimal polynomial

$-9 = (-1)(3)(\lambda_3) \Rightarrow \lambda_3 = 3$

$tr(A) = -1 + 3 + 3 = 5$

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
End-semester Examination
PART-A
Date: December 14, 2020. Time: 70 minutes Max. marks: 30

Answer all questions. Each question carries 5 marks.

1. Let V be a vector space over $\mathbb{F}$, $w \in V$ and $\{v_1, v_2, \ldots, v_n\}$ be a linearly independent subset of V. Prove that if $\{v_1 + w, v_2 + w, \ldots, v_n + w\}$ is a linearly dependent set, then $w \in Span\{v_1, v_2, \ldots, v_n\}$. Is the converse true? Justify.

2. Suppose that P, Q and R are nontrivial subspaces of a vector space V such that $P + Q = P + R$. Should $Q = R$ necessarily? Justify your conclusion.

3. Let S denote the subset $\{(1, -1, 0), (0, -3, 2), (2, 1, -2), (3, 3, -4)\}$ of $\mathbb{R}^3$. Find the smallest set $R \subset S$ such that $Span(R) = Span(S)$.

4. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c, d are real numbers. For $x, y \in \mathbb{R}^2$, let $f_A(x, y) = y^T Ax$ be an inner product on $\mathbb{R}^2$. Show that $b = c$, $a > 0$, $d > 0$, and $ad - bc > 0$.

5. Let V and W be finite dimensional vector spaces over $\mathbb{F}$ and $T : V \to W$ is a linear map. When is T said to be an isomorphism? Prove that T is an isomorphism if and only if $\{T(v_1), T(v_2), \ldots, T(v_n)\}$ is a basis for W whenever $\{v_1, v_2, \ldots, v_n\}$ is a basis for V.

6. Let V and W be finite dimensional inner product spaces over the same field. Let $T : V \longrightarrow W$ and $S : W \longrightarrow V$ be linear transformations. Suppose that $TS = 0$. Then show the following:

   (a) $\dim V \geq rank(T) + rank(S)$.

   (b) $\dim W \leq rank(T^*) + rank(S^*)$.

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Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
End-semester Examination
PART-B
Date: December 14, 2020. Time: 70 minutes Max. marks: 30

Answer all questions. Each question carries 5 marks.

1. Let $V = P_2(\mathbb{R})$ be the vector space of all polynomials of degree not exceeding 2. Let $\mathcal{B} = \{1 - 3t^2, 2 + t - 5t^2, 1 + 2t\}$ and $\mathcal{C} = \{1, t, t^2\}$ be ordered bases for V. Suppose that $T : V \to V$ is the differentiation map.
   (a) Compute $[T]_{\mathcal{C}}$.
   (b) Let A denote the change of coordinates matrix that changes the $\mathcal{C}$ coordinates of an element of V to its $\mathcal{B}$ coordinates. Compute the matrix A.
   (c) Express $[T]_{\mathcal{B}}$ in terms of the matrices A and $[T]_{\mathcal{C}}$.

2. Let $V = P_2(\mathbb{R})$ be the space of all polynomials of degree not exceeding 2, endowed with the inner product $\langle f(t), g(t) \rangle = \int_0^1 f(t)g(t)\,dt$. Suppose that $T : V \to V$ is the linear map defined by $T(a_0 + a_1 t + a_2 t^2) = a_1 t$.
   (a) Demonstrate that T is not self-adjoint by considering $\langle T(1), t \rangle$.
   (b) Let $\mathcal{C} = \{1, t, t^2\}$ and $B = [T]_{\mathcal{C}}$. Show that $B^* = B$.
   (c) Although $B^* = B$, T is not self-adjoint as observed in part (a). Why is this not a contradiction? Justify.

3. Let $V = P(\mathbb{R})$ denote the inner product space of all polynomials with real coefficients, endowed with the inner product $\langle f(t), g(t) \rangle = \int_0^1 f(t)g(t)dt$, for $f, g \in V$. Let $W = P_1$ denote the subspace of polynomials of degree less than or equal to 1. Find the best approximation to $t^2 \in V$ from the subspace W.

4. Let V be an n-dimensional inner product space over $\mathbb{R}$ and $T : V \to V$ be a self-adjoint linear map.

   (a) Prove that eigenvalues corresponding to distinct eigenvalues of T are orthogonal.

   (b) Suppose that $dim(Range(T)) = k < n$. What is the maximum number of distinct eigenvalues T can have? Justify your conclusion.

5. Let $A \in M_3(\mathbb{R})$ denote the matrix all of whose entries are one. Find all the eigenvalues of A and obtain two non-trivial subspaces W of $\mathbb{R}^3$ such that $Ax \in W$ for all $x \in W$.

6. Let V be a finite dimensional vector space over $\mathbb{C}$ and $T : V \to V$ be a linear map.

   (a) State the Cayley-Hamilton theorem for the linear map T.

   (b) The linear map $T : V \to V$ is said to be nilpotent if $T^k = 0$ for some $k \in \mathbb{N}$. Using part (a) or otherwise, prove that $T : V \to V$ is nilpotent if and only if the only eigenvalue of T is 0.

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Indian Institute of Technology Madras

Department of Mathematics

MA 1102 – Series and Matrices

End Semester Examination

Date: 07.05.2024

Time: 09:00 – 12:00

Class: I B. Tech.

Maximum Marks: 60

Write your roll number and name on both **PART A** and **PART B** answer books.
Write your teacher's name on the top right corner of both the answer books.
Use of any electronic gadgets is not allowed during this examination.

PART A

(1) (a) Does the series $\sum_{n=1}^{\infty} \sin\frac{1}{n}$ converge? Justify. [2]

(b) Determine the values of s for which the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^s}$ converges. [3]

(2) Test the convergence of $\int_1^{\infty} \frac{\sin x}{x^p}\, dx$ for $p > 0$. [5]

(3) Using the Taylor series of $f(t) = e^{-t^4}$ around the point $t = 0$, find the power series expansion of the function $F(x) = \int_0^x e^{-t^4}\, dt$. [4]

(4) Let $f(x) = e^{x^{10}} + e^{-x^{10}} + 2\cos(x^{10}) - 4$. Find the smallest natural number n such that $f^{(n)}(0) \neq 0$ and also find the value of this $f^{(n)}(0)$. [4]

(5) Find the half-range cosine series for $f(x) = x + 1$, $0 < x < 1$, and show that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$. [5]

(6) (a) If $A \in \mathbb{R}^{2\times 1}$ and $B \in \mathbb{R}^{1\times 2}$, then prove that the matrix AB is not invertible. [2]

(b) Show that if a real matrix is orthogonal and triangular, then it must be diagonal. Determine the diagonal entries of such a matrix. [4]

[P.T.O.]

$A \cdot A^T = I$

$A = \left[\begin{array}{c|c} P & P_1 \\ \hline 0 & U \end{array}\right]$

$\begin{bmatrix} P^2 + P_1 \cdot P_1 & P_1 \cdot U \\ U \cdot P_1 & U^2 \end{bmatrix}$

$A^T = \left[\begin{array}{c|c} P & 0 \\ \hline P_1 & U \end{array}\right]$

$P_1 = 0$

$U^2 \rightarrow$ diagonal

$P = 1$

(6) (a) If $A \in \mathbb{R}^{2\times 1}$ and $B \in \mathbb{R}^{1\times 2}$, then prove that the matrix AB is not invertible. [2]
(b) Show that if a real matrix is orthogonal and triangular, then it must be diagonal. Determine the diagonal entries of such a matrix. [4]

(a) $A = [a\ b]^t \quad B = [c\ d]$

$\Rightarrow AB = \begin{bmatrix} ac & ad \\ bc & bd \end{bmatrix} \Rightarrow |AB| = ac.bd - ad.bc = 0$

(b) $A.A^T = A^TA = I$

orthogonal matrices have orthonormal columns

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \cdots & a_{1n} \\ 0 & a_{22} & a_{23} \cdots & a_{2n} \\ 0 & 0 & a_{33} \cdots & a_{3n} \\ 0 & 0 & 0 \cdots & a_{nn} \end{bmatrix}$$

$\Rightarrow a_{11}.a_{12} = 0$ for any a_{11}

$\Rightarrow a_{12} = 0$

$a_{12}.a_{13} + a_{22}.a_{23} = 0$

$\Rightarrow a_{23} = 0$

all non diagonal elements come out to be 0

(7) Apply the Gram-Schmidt orthogonalization process to the vectors $u_1 = (2, 2, 0, 2)$, $u_2 = (0, 1, 1, -1)$, $u_3 = (1, 0, 1, 0)$ and $u_4 = (2, 4, 2, 1)$. [4]

$v_1 = [2, 2, 0, 2]$

$v_2 = u_2 - \frac{\langle v_1 . u_2 \rangle}{\langle v_1 . v_1 \rangle} v_1$

$= [0, 1, 1, -1]$

$v_3 = u_3 - \frac{\langle v_1 u_3 \rangle}{\langle v_1 . v_1 \rangle} v_1 - \frac{\langle v_2 u_3 \rangle}{\langle v_2 . v_2 \rangle} v_2$

$= [1, 0, 1, 0] - \frac{2}{12}[2, 2, 0, 2] - \frac{1}{3}[0, 1, 1, -1]$

$= \left[\frac{2}{3}, -\frac{2}{3}, \frac{2}{3}, 0\right]$

$$v_4 = u_4 - \frac{\langle v_1, u_4\rangle}{\langle v_1, v_1\rangle} v_1 - \frac{\langle v_2, u_4\rangle}{\langle v_2, v_2\rangle} v_2 - \frac{\langle v_3, u_4\rangle}{\langle v_3, v_3\rangle} v_3$$

$$= [2,4,2,1] - \frac{14}{12}[2,2,0,2] - \frac{5}{3}[0,1,1,-1] - 0$$

$$= [-1/3, 0, 1/3, 1/3]$$

(12) Using the Cayley-Hamilton theorem, find the inverse of the matrix [4]

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 1 & 2 \\ -1 & 3 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 9 & -1 & 5 \\ -6 & 8 & 9 \\ -10 & 16 & 21 \end{bmatrix}$$

$$\begin{bmatrix} \lambda-3 & 1 & -1 \\ 1 & \lambda-1 & -2 \\ 1 & -3 & \lambda-4 \end{bmatrix} = (\lambda-3)[\lambda^2-5\lambda+4-6] - 1(\lambda-4+2) - 1(-3-\lambda+1)$$

$$= \lambda^3 - 8\lambda^2 + 13\lambda + 6 - \lambda + 2 + \lambda + 2$$

$$= \lambda^3 - 8\lambda^2 + 13\lambda + 10 = 0$$

$$A^3 - 8A^2 + 13A + 10I = 0$$

$$-\frac{(A^2 - 8A + 13I)}{10} = A^{-1} = -\frac{\left[\begin{bmatrix} 9 & -1 & 5 \\ -6 & 8 & 9 \\ -10 & 16 & 21 \end{bmatrix} - \begin{bmatrix} 24 & -8 & 8 \\ -8 & 8 & 16 \\ -8 & 24 & 32 \end{bmatrix} + \begin{bmatrix} 13 & 0 & 0 \\ 0 & 13 & 0 \\ 0 & 0 & 13 \end{bmatrix}\right]}{10}$$

$$\Rightarrow A^{-1} = \frac{1}{10}\begin{bmatrix} 2 & -7 & 3 \\ -2 & -13 & 7 \\ 2 & 8 & -2 \end{bmatrix}$$

PART B

(7) Apply the Gram-Schmidt orthogonalization process to the vectors $u_1 = (2, 2, 0, 2)$, $u_2 = (0, 1, 1, -1)$, $u_3 = (1, 0, 1, 0)$ and $u_4 = (2, 4, 2, 1)$. [4]

(8) Let $A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 3 & 5 & 7 & 1 \\ 1 & 5 & 4 & 5 \\ 2 & 8 & 7 & 9 \end{bmatrix}$.

(a) Find the RREF of A.
(b) Find the rank of A.
(c) Express one of the columns as a linear combination of the other columns. [6]

(9) Let $A \in \mathbb{R}^{3\times 3}$ be such that

$$A\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}; \quad A\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}; \quad \text{and} \quad A\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}.$$

(a) Find all eigenvalues of A. (b) Show that A is diagonalizable.
(c) Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. [5]

(10) Consider the system of linear equations

$$5x + 3y + 8z = 5$$
$$3x + 2y + 5z = 6$$
$$px + 4y + 4z = q.$$

Find conditions on p and q so that the given system of linear equations has
(a) no solution; (b) unique solution; (c) infinitely many solutions. [7]

(11) Find all eigenvalues and three linearly independent eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}.$$ [5]

(12) Using the Cayley-Hamilton theorem, find the inverse of the matrix [4]

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 1 & 2 \\ -1 & 3 & 4 \end{bmatrix}.$$

(11) Find all eigenvalues and three linearly independent eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}.$$ [5]

$$\begin{vmatrix} \lambda-4 & 0 & -1 \\ -2 & \lambda-3 & -2 \\ -1 & 0 & \lambda-4 \end{vmatrix} \quad (\lambda-4)[\lambda^2-7\lambda+12] - 1(\lambda-3)$$

$$= (\lambda-3)^2(\lambda-5)$$

$$\Leftarrow \sigma(A) = \{3, 3, 5\}$$

$$4v_1 + v_3 = 3v_1 \Rightarrow v_1 = -v_3$$

$$2v_1 + 3v_2 + 2v_3 = 3v_2 \Rightarrow \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v_1 + 4v_3 = 3v_3$$

$$4v_1 + v_3 = 5v_1 \Rightarrow v_1 = v_3$$

$$2v_1 + 3v_2 + 2v_3 = 5v_2 \Rightarrow v_2 = 2v_1 \Rightarrow \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$v_1 + 4v_3 = 5v_3$$

(9) Let $A \in \mathbb{R}^{3\times 3}$ be such that

$$A\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}; \quad A\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}; \quad \text{and} \quad A\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}.$$

(a) Find all eigenvalues of A. (b) Show that A is diagonalizable.

(c) Find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. [5]

$\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 0$ $A\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + A\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}$

(a) 0, 2, 3

(b) $[1,2,0]^t, [1,0,2]^t, [0,-1,0]^t \Rightarrow$ forms a basis

$\Rightarrow A$ is diagonalizable

(c) $D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ $P = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 0 & 2 & 0 \end{bmatrix}$

(10) Consider the system of linear equations

$$5x + 3y + 8z = 5$$
$$3x + 2y + 5z = 6$$
$$px + 4y + 4z = q.$$

Find conditions on p and q so that the given system of linear equations has

(a) no solution; (b) unique solution; (c) infinitely many solutions. [7]

$$\begin{bmatrix} 5 & 3 & 8 & 5 \\ 3 & 2 & 5 & 6 \\ p & 4 & 4 & q \end{bmatrix}$$

$\downarrow E_{1/5}[1]$

$$\begin{bmatrix} 1 & 3/5 & 8/5 & 1 \\ 3 & 2 & 5 & 6 \\ p & 4 & 4 & q \end{bmatrix} \xrightarrow{\substack{E_{-3}[2,1] \\ E_{-p}[3,1]}} \begin{bmatrix} 1 & 3/5 & 8/5 & 1 \\ 0 & 1/5 & 1/5 & 3 \\ 0 & \frac{20-3p}{5} & \frac{20-8p}{5} & q-p \end{bmatrix}$$

$\downarrow = E_{-3}[1,2]$

$E_{3p-20}[3,2]$

$$\begin{bmatrix} 1 & 0 & 1 & -8 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -p & 8p+q-60 \end{bmatrix} \xleftarrow{E_5[2]} \begin{bmatrix} 1 & 0 & 1 & -8 \\ 0 & 1/5 & 1/5 & 3 \\ 0 & 0 & -p & 8p+q-60 \end{bmatrix}$$

$\downarrow E_{-1/p}[3]$ $\boxed{p \neq 0}$

$$\begin{bmatrix} 1 & 0 & 1 & -8 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & \frac{60-8p-q}{p} \end{bmatrix} \xrightarrow[E_{-1}[2,3]]{E_{-1}[1,3]} \begin{bmatrix} 1 & 0 & 0 & \frac{q-60}{p} \\ 0 & 1 & 0 & \frac{11p+q-60}{p} \\ 0 & 0 & 1 & \frac{60-8p-q}{p} \end{bmatrix}$$

- No Solution: $\text{rank}(A) < \text{rank}(A|B) \Rightarrow$ not possible here under $p \neq 0 \Rightarrow p = 0$

  But if $p = 0 \rightarrow 3^{rd}$ row becomes $[0, 0, 0, q-60]$

  $\Rightarrow$ No Solution $\Rightarrow q - 60 \neq 0 \Rightarrow p = 0, q \neq 60$

- one Solution: $\text{rank}(A) = \text{rank}(A|B) = 3 \Rightarrow p \neq 0$

- ∞ Solution: $\text{rank}(A) = \text{rank}(A|B) < 3 \Rightarrow$ not possible under $p \neq 0 \Rightarrow p = 0, q = 60$

AKSHITH · 4

Indian Institute of Technology Madras
Department of Mathematics
MA 1102 : Series and Matrices (End Semester Exam)

Max.Marks : 60 | Duration : 3 hours
Date : 26-6-2023 | Time : 9:00-12:00 hrs

(1) Write your **name, roll number** and **Teacher's name** on the first page of answer script.
(2) Use **separate answer sheets** for Part A and Part B.
(3) Use of electronic gadgets is **NOT** permitted.

PART - A

1. Find the power series (in powers of x) of $\dfrac{\log(1+x)}{(1-x)^2}$, $|x| < 1$ [3]

2. Using Taylor's theorem (Taylor's formula) show that

$$\left|\sqrt{1+x} - \left(1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}\right)\right| \leq x^4 \quad \text{for } x > 0.$$

[4]

3. Determine the values of x for which the series $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n (x+1)^n}{(n^2+1)}$ converges . [4]

4. Construct the half-range Fourier sine series for $f(x) = 3 + x$ in $0 \leq x < 3$. Hence

   (i) find the sum of the half range Fourier sine series at $x = 0$. Justify your result.

   (ii) compute the value of $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$.

[5]

5. For what value of a and b, the rank of matrix A is 2.

$$A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 2 \\ 6 & -2 & a & b \end{bmatrix}$$

[3]

6. Suppose that the vectors $\{u, v, w\}$ are linearly independent. Show that the vectors $\{u+v, u-v, u-2v+w\}$ are also linearly independent. [3]

7. Find the row reduced echelon form (RREF) of the matrix

$$A = \begin{bmatrix} 3 & -3 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & -2 \\ -1 & 1 & 2 \end{bmatrix}$$

[4]

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PART - B

8. Let $v_1 = (1,3,4)$, $v_2 = (-4,2,-6)$, $v_3 = (-3,-2,c)$. Find the value of c such that v_1, v_2, v_3 are linearly dependent. Using the obtained value of c, find α, β, γ such that $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$. [5]

9. Let $u_1 = (1,2,1)$, $u_2 = (1,1,3)$, $u_3 = (2,1,1)$. Apply Gram-Schmidt process to u_1, u_2, u_3 and construct orthonormal vectors v_1, v_2, v_3. [4]

10. Using Gauss-Jordan elimination method, determine the values of $k \in \mathbb{R}$ so that the system of linear equations

$$x + y + z = 1,$$
$$2x + 3y + kz = 3,$$
$$x + ky + 3z = 3$$

has (a) no solution (b) infinitely many solutions (c) exactly one solution. Also, find solutions of the given system, if it exists. [8]

11. Let A be 6×6 real non-invertible matrix with each diagonal element is equal to 2. If $(4+i)$ and $(4-3i)$ are two eigenvalues of A, then find all other eigenvalues of A. [4]

12. Let A be $n \times n$ real symmetric matrix.

(a) Prove that every eigenvalue of A is real.

(b) Let λ_1 and λ_2 be two distinct eigenvalues of A with corresponding eigenvectors v_1 and v_2. Prove that v_1 and v_2 are orthogonal. [2+3]

13. Show that if $n \times n$ matrix A has n linearly independent eigenvectors, then A^T also has n linearly independent eigenvectors. [3]

14. Let

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}.$$

Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$ [5]

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